

2 PRU – 1PUS Parallel Manipulator:

1.KINEMATICS PART:

Given Motion Inputs

The moving platform of the 2PRU–1PUS parallel manipulator undergoes the following motions:

- Vertical translation along the z -axis:

$$Z = a_1 \sin(\omega t)$$

- Roll rotation about the x -axis:

$$\phi = a_2 \sin(\omega t)$$

- Pitch rotation about the y -axis:

$$\theta = a_3 \sin(\omega t)$$

Base Slider Coordinates

The coordinates of the base sliders in the fixed global frame are defined as:

$$S_1 = (S_B, 0, Z_1)$$

$$S_2 = (-S_B, 0, Z_2)$$

$$S_3 = (0, S_B, Z_3)$$

Each slider is constrained to move only along the vertical direction.

Platform Attachment Points (Local Frame)

In the local coordinate frame attached to the moving platform, the attachment points are defined as:

$$P'_1 = (S_m, 0, 0)$$

$$P'_2 = (-S_m, 0, 0)$$

$$P'_3 = (0, S_m, 0)$$

Rotation Modeling

The orientation of the platform is represented using successive rotations about the y - and x -axes. The overall rotation matrix is given by:

$$R = R_y(\theta)R_x(\phi)$$

where,

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R = \begin{bmatrix} \cos \theta & \sin \theta \sin \phi & \sin \theta \cos \phi \\ 0 & \cos \phi & -\sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix}$$

Rigid Body Transformation

The global coordinates of the platform attachment points are obtained using rigid body transformation:

$$P_i = RP'_i + d$$

where the translation vector is:

$$d = \begin{bmatrix} 0 \\ 0 \\ Z \end{bmatrix}$$

Global Coordinates of Platform Points

After applying the above transformation, the platform points are obtained as:

$$P_1 = \begin{bmatrix} S_m \cos \theta \\ 0 \\ Z - S_m \sin \theta \end{bmatrix}$$

$$P_2 = \begin{bmatrix} -S_m \cos \theta \\ 0 \\ Z + S_m \sin \theta \end{bmatrix}$$

$$P_3 = \begin{bmatrix} S_m \sin \theta \sin \phi \\ S_m \cos \phi \\ Z + S_m \cos \theta \sin \phi \end{bmatrix}$$

Leg Length Constraint

Each leg of the manipulator has a fixed length L . Therefore, the distance between each platform point and its corresponding base slider satisfies:

$$\|P_i - S_i\|^2 = L^2$$

Inverse Kinematics for Slider Z_1

Applying the constraint to leg $P_1 - S_1$:

$$(P_{1x} - S_{1x})^2 + (P_{1z} - Z_1)^2 = L^2$$

Substituting coordinates:

$$(S_m \cos \theta - S_B)^2 + (Z - S_m \sin \theta - Z_1)^2 = L^2$$

Define:

$$a = S_B - S_m \cos \theta, \quad b = S_m \sin \theta$$

Solving for Z_1 :

$$Z_1 = Z - S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2}$$

Inverse Kinematics for Slider Z_2

Similarly, for leg $P_2 - S_2$:

$$Z_2 = Z + S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2}$$

Inverse Kinematics for Slider Z_3

For leg $P_3 - S_3$, both x - and y -direction offsets exist:

$$(S_m \sin \theta \sin \phi)^2 + (S_m \cos \phi - S_B)^2 + (Z + S_m \cos \theta \sin \phi - Z_3)^2 = L^2$$

Define:

$$\begin{aligned} c &= S_m \sin \theta \sin \phi \\ d &= S_m \cos \phi - S_B \\ e &= S_m \cos \theta \sin \phi \end{aligned}$$

Solving for Z_3 :

$$Z_3 = Z + S_m \cos \theta \sin \phi - \sqrt{L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_B)^2}$$

Final Inverse Kinematic Solution

$$\begin{aligned} Z_1 &= Z - S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2} \\ Z_2 &= Z + S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2} \\ Z_3 &= Z + S_m \cos \theta \sin \phi - \sqrt{L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_B)^2} \end{aligned}$$

For given Values :

Fixed geometry and inputs

All values in **mm**

- $S_m = 70$
- $S_B = 75$
- $S_{B3} = 70.7$

- $L = 25$

Platform inputs

- Heave:

$$Z = 50 \text{ mm}$$

- Pitch: θ

- Roll: ϕ

Final inverse kinematics equations (with $Z = 50$ substituted)

These are your **final working equations**.

◆ Slider 1

$$Z_1 = 50 - 70 \sin \theta - \sqrt{25^2 - (75 - 70 \cos \theta)^2}$$

◆ Slider 2

$$Z_2 = 50 + 70 \sin \theta - \sqrt{25^2 - (75 - 70 \cos \theta)^2}$$

◆ Slider 3

$$Z_3 = 50 + 70 \cos \theta \sin \phi - \sqrt{25^2 - (70 \sin \theta \sin \phi)^2 - (70 \cos \phi - 70.7)^2}$$

Numerical check at neutral orientation

Case:

$$\theta = 0, \phi = 0$$

Compute common terms

$$\cos 0 = 1, \sin 0 = 0$$

◆ Z_1 and Z_2

$$75 - 70 \cos 0 = 75 - 70 = 5$$

$$\sqrt{25^2 - 5^2} = \sqrt{625 - 25} = \sqrt{600} \approx 24.495$$

$$Z_1 = Z_2 = 50 - 24.495 = 25.505 \text{ mm}$$

◆ Z_3

$$70 \cos 0 - 70.7 = -0.7$$

$$\sqrt{25^2 - 0.7^2} = \sqrt{625 - 0.49} \approx 24.99$$

$$Z_3 = 50 - 24.99 = 25.01 \text{ mm}$$

Final numerical result ($\theta = 0, \phi = 0$)

$$\begin{aligned} Z_1 &= 25.51 \text{ mm} \\ Z_2 &= 25.51 \text{ mm} \\ Z_3 &= 25.01 \text{ mm} \end{aligned}$$

Z_3 is slightly different?

Because:

$$S_{B3} \neq S_B$$

For a nominal platform height of 50 mm and zero pitch and roll angles, the inverse kinematics yields prismatic joint displacements of approximately 25.5 mm for sliders 1 and 2, and 25.0 mm for slider 3. The slight difference in the third actuator displacement arises from the asymmetric base geometry of the third leg.

The motion is valid **only if**:

$$\begin{aligned} 25^2 - (75 - 70 \cos \theta)^2 &\geq 0 \\ 25^2 - (70 \sin \theta \sin \phi)^2 - (70 \cos \phi - 70.7)^2 &\geq 0 \end{aligned}$$

These inequalities define the **coverage / workspace**.

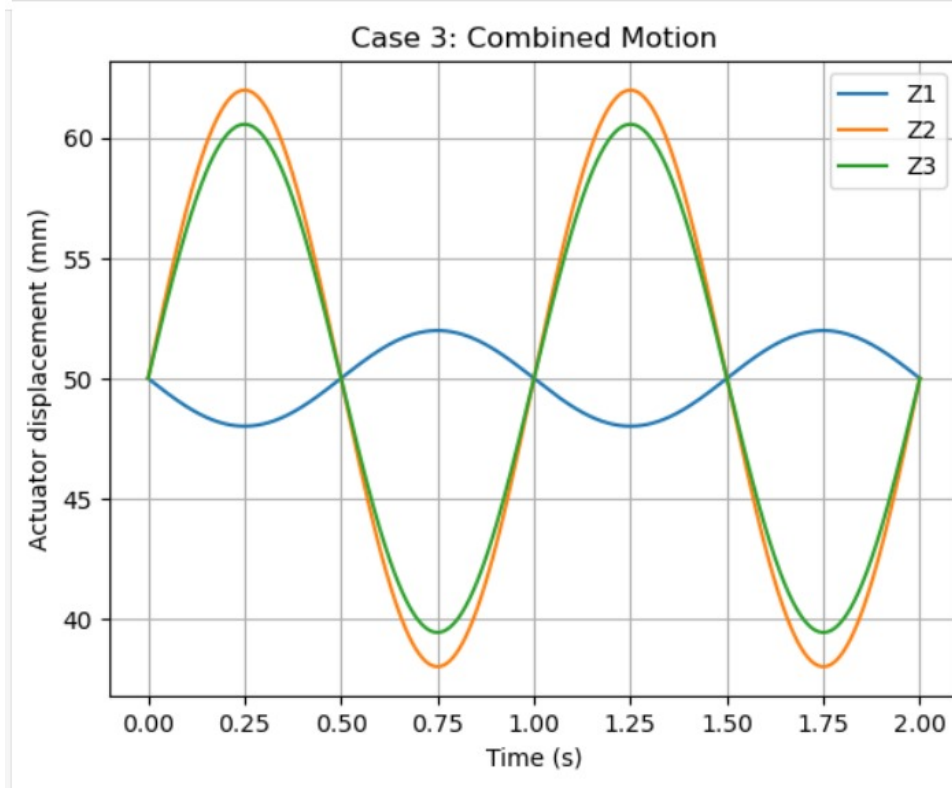


Figure 1: Actuator displacements (Z_1, Z_2, Z_3) for Case 3: Combined Motion.

Types of Singularities:

1. Inverse Kinematic Singularities

When small platform motion $\rightarrow$ infinite change in Z_1, Z_2, Z_3

Related to actuator alignment

2. Direct Kinematic Singularities

When actuators move but platform pose is not uniquely defined

Very dangerous

3. Constraint Singularities

When constraint forces become dependent

Platform loses resistance to external forces

Mathematical Tool: Jacobian Matrix

Singularity analysis is done using the Jacobian.

For your system:

$$\begin{bmatrix} \dot{Z}_1 \\ \dot{Z}_2 \\ \dot{Z}_3 \end{bmatrix} = \mathbf{J} \begin{bmatrix} \dot{z} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

Where:

$\mathbf{J}$ = Jacobian matrix

Singularity occurs when:

$$\det(\mathbf{J}) = 0$$

What Jacobian We Need

We already have the inverse kinematics for our three legs: $Z_1(z, \theta, \phi)$, $Z_2(z, \theta, \phi)$, and $Z_3(z, \theta, \phi)$.

Now we differentiate them with respect to time to get the velocity relationship:

$$\begin{bmatrix} \dot{Z}_1 \\ \dot{Z}_2 \\ \dot{Z}_3 \end{bmatrix} = \mathbf{J} \begin{bmatrix} \dot{z} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

Where the Jacobian matrix $\mathbf{J}$ is defined as the matrix of partial derivatives:

$$\mathbf{J} = \frac{\partial(Z_1, Z_2, Z_3)}{\partial(z, \theta, \phi)}$$

Write Final Position Equations Clearly

From our final verified expressions:

Leg 1

$$Z_1 = z - S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2}$$

Leg 2

$$Z_2 = z + S_m \sin \theta - \sqrt{L^2 - (S_B - S_m \cos \theta)^2}$$

Leg 3

$$Z_3 = z + S_m \cos \theta \sin \phi - \sqrt{L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2}$$

Compute Partial Derivatives

We now compute $\frac{\partial Z_i}{\partial z}$, $\frac{\partial Z_i}{\partial \theta}$, and $\frac{\partial Z_i}{\partial \phi}$ carefully.

Leg 1 Derivatives

$$\partial Z_1 / \partial z$$

$$\frac{\partial Z_1}{\partial z} = 1$$

(Simple. Linear.)

$$\partial Z_1 / \partial \theta$$

Break it down carefully:

$$Z_1 = z - S_m \sin \theta - \sqrt{F(\theta)}$$

Where:

$$F(\theta) = L^2 - (S_B - S_m \cos \theta)^2$$

Derivative of the square root:

$$\frac{d}{d\theta} \sqrt{F} = \frac{1}{2\sqrt{F}} F'$$

Now compute F' :

$$F' = 2(S_B - S_m \cos \theta)(S_m \sin \theta)$$

So:

$$\frac{\partial Z_1}{\partial \theta} = -S_m \cos \theta - \frac{(S_B - S_m \cos \theta)(S_m \sin \theta)}{\sqrt{L^2 - (S_B - S_m \cos \theta)^2}}$$

$$\partial Z_1 / \partial \phi$$

$$\frac{\partial Z_1}{\partial \phi} = 0$$

Leg 1 is unaffected by roll.

Leg 2 Derivatives

Very similar to Leg 1.

$$\partial Z_2 / \partial z$$

$$\frac{\partial Z_2}{\partial z} = 1$$

$$\partial Z_2 / \partial \theta$$

$$\frac{\partial Z_2}{\partial \theta} = +S_m \cos \theta - \frac{(S_B - S_m \cos \theta)(S_m \sin \theta)}{\sqrt{L^2 - (S_B - S_m \cos \theta)^2}}$$

(Sign change in the first term only.)

$$\partial Z_2 / \partial \phi$$

$$\frac{\partial Z_2}{\partial \phi} = 0$$

Leg 3 Derivatives

This one is more complex. Let:

$$G = L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2$$

$\partial Z_3 / \partial z$

$$\frac{\partial Z_3}{\partial z} = 1$$

$\partial Z_3 / \partial \theta$

There are two contributions. First term:

$$\frac{d}{d\theta}(S_m \cos \theta \sin \phi) = -S_m \sin \theta \sin \phi$$

Second term:

$$\frac{1}{2\sqrt{G}} \cdot 2(S_m \sin \theta \sin \phi)(S_m \cos \theta \sin \phi)$$

So:

$$\frac{\partial Z_3}{\partial \theta} = -S_m \sin \theta \sin \phi - \frac{S_m^2 \sin \theta \cos \theta \sin^2 \phi}{\sqrt{G}}$$

$\partial Z_3 / \partial \phi$

First term derivative:

$$S_m \cos \theta \cos \phi$$

Second term derivative gives:

$$-\frac{S_m^2 \sin^2 \theta \sin \phi \cos \phi + (S_m \cos \phi - S_{B3})(S_m \sin \phi)}{\sqrt{G}}$$

So:

$$\frac{\partial Z_3}{\partial \phi} = S_m \cos \theta \cos \phi - \frac{S_m^2 \sin^2 \theta \sin \phi \cos \phi + (S_m \cos \phi - S_{B3})(S_m \sin \phi)}{\sqrt{G}}$$

— Final Jacobian Matrix

Combining everything, our Jacobian matrix is:

$$\mathbf{J} = \begin{bmatrix} 1 & J_{12} & 0 \\ 1 & J_{22} & 0 \\ 1 & J_{32} & J_{33} \end{bmatrix}$$

Where the detailed terms ($J_{12}, J_{22}, J_{32}, J_{33}$) are exactly the partial derivatives computed above.

— Singularity Condition

For a singularity to occur:

$$\boxed{\det(\mathbf{J}) = 0}$$

Since the first column consists entirely of 1's, the determinant simplifies nicely. We will compute that next.

— Write Jacobian in Structured Form

From the previous derivation:

$$\mathbf{J} = \begin{bmatrix} 1 & J_{12} & 0 \\ 1 & J_{22} & 0 \\ 1 & J_{32} & J_{33} \end{bmatrix}$$

Where:

$$J_{12} = \frac{\partial Z_1}{\partial \theta}, \quad J_{22} = \frac{\partial Z_2}{\partial \theta}, \quad J_{32} = \frac{\partial Z_3}{\partial \theta}, \quad J_{33} = \frac{\partial Z_3}{\partial \phi}$$

Notice the important structure:

- First column = all 1's
- Third column = only one non-zero term (J_{33})

This simplifies the determinant dramatically.

— Compute Determinant

$$\det(\mathbf{J}) = \begin{vmatrix} 1 & J_{12} & 0 \\ 1 & J_{22} & 0 \\ 1 & J_{32} & J_{33} \end{vmatrix}$$

Expand along the third column (the best choice, as the only element contributing is J_{33}):

$$\det(\mathbf{J}) = J_{33} \cdot \begin{vmatrix} 1 & J_{12} \\ 1 & J_{22} \end{vmatrix}$$

Now compute the 2×2 determinant:

$$\det(\mathbf{J}) = J_{33}(1 \cdot J_{22} - 1 \cdot J_{12})$$

So the final expression is:

$$\det(\mathbf{J}) = J_{33}(J_{22} - J_{12})$$

— First Singularity Condition

$$J_{33} = 0$$

This means:

$$\frac{\partial Z_3}{\partial \phi} = 0$$

Physical meaning:

- Leg 3 no longer contributes independent roll control.
- Platform loses one rotational DOF.
- Roll becomes uncontrollable.
- This is a Type-1 singularity.

— Second Singularity Condition

$$\boxed{J_{22} - J_{12} = 0}$$

Recall the partial derivatives:

$$J_{12} = -S_m \cos \theta - \frac{(S_B - S_m \cos \theta)(S_m \sin \theta)}{\sqrt{L^2 - (S_B - S_m \cos \theta)^2}}$$
$$J_{22} = +S_m \cos \theta - \frac{(S_B - S_m \cos \theta)(S_m \sin \theta)}{\sqrt{L^2 - (S_B - S_m \cos \theta)^2}}$$

Subtract them:

$$J_{22} - J_{12} = 2S_m \cos \theta$$

A beautiful simplification. So the second condition becomes:

$$\boxed{2S_m \cos \theta = 0}$$

Since $S_m \neq 0$:

$$\boxed{\cos \theta = 0} \implies \boxed{\theta = \pm \frac{\pi}{2}}$$

Final Singularity Conditions

Combining the conditions, the determinant is zero when:

$$\det(\mathbf{J}) = 0 \implies \begin{cases} \cos \theta = 0 \\ \text{OR} \\ \frac{\partial Z_3}{\partial \phi} = 0 \end{cases}$$

Physical Interpretation

Singularity 1 — $\theta = \pm 90^\circ$

The platform becomes vertical, and Legs 1 and 2 align. There is a loss of pitch controllability and the mechanism collapses sideways. This is a geometric singularity.

Singularity 2 — $J_{33} = 0$

Occurs when $\frac{\partial Z_3}{\partial \phi} = 0$. Physically, Leg 3 becomes aligned with the roll axis. Roll motion produces no actuator change, resulting in a loss of roll control.

Most Important Hidden Singularity

Also remember the denominator in the Jacobian terms:

$$\sqrt{L^2 - (S_B - S_m \cos \theta)^2}$$

If this becomes zero:

$$L^2 - (S_B - S_m \cos \theta)^2 = 0$$

Then the leg becomes horizontal. The vertical stiffness becomes 0, and an infinite actuator velocity is required to move the platform. This represents another distinct singularity surface.

The manipulator exhibits singular configurations when either the pitch angle satisfies $\theta = \pm 90^\circ$, causing geometric alignment of the first two legs, or when the roll Jacobian term vanishes, resulting in loss of independent roll control. Additional singularities arise when the horizontal projection of a leg equals its length, causing the Jacobian denominator to approach zero.

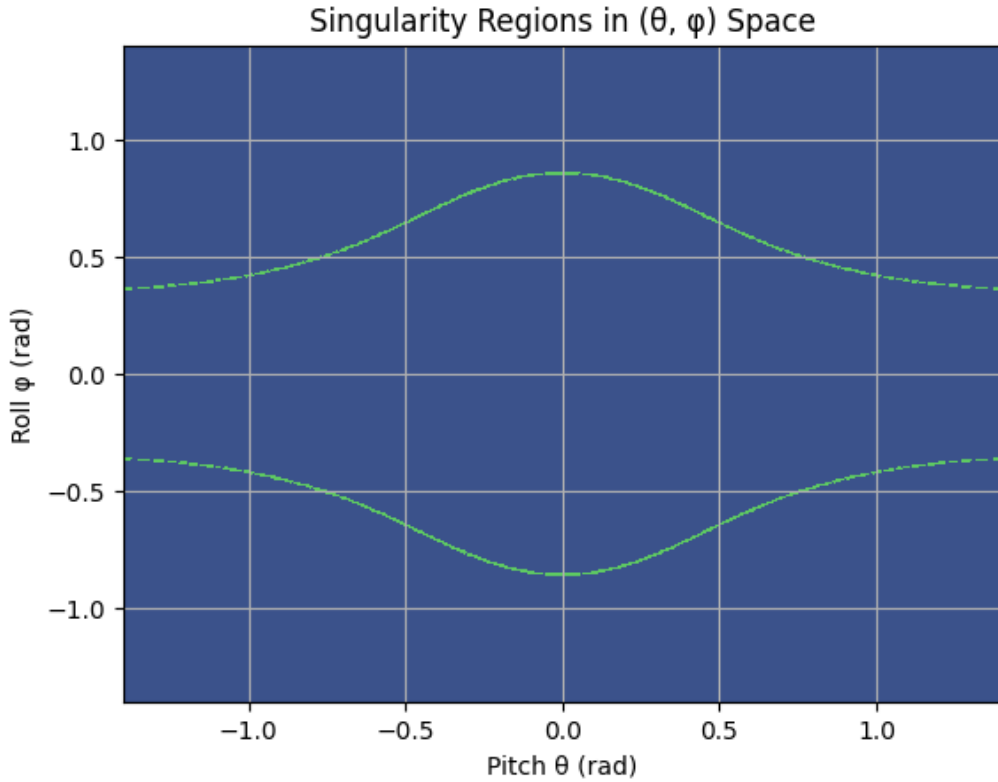


Figure 2: Actuator displacements (Z_1, Z_2, Z_3) for Case 3: Combined Motion.

First Plot — Singularity Region

This shows configurations where:

$$\det(\mathbf{J}) = 0$$

Meaning:

- Loss of control
- Infinite actuator velocity
- Mechanism stiffness collapse

These boundary curves correspond to:

$$L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2 = 0$$

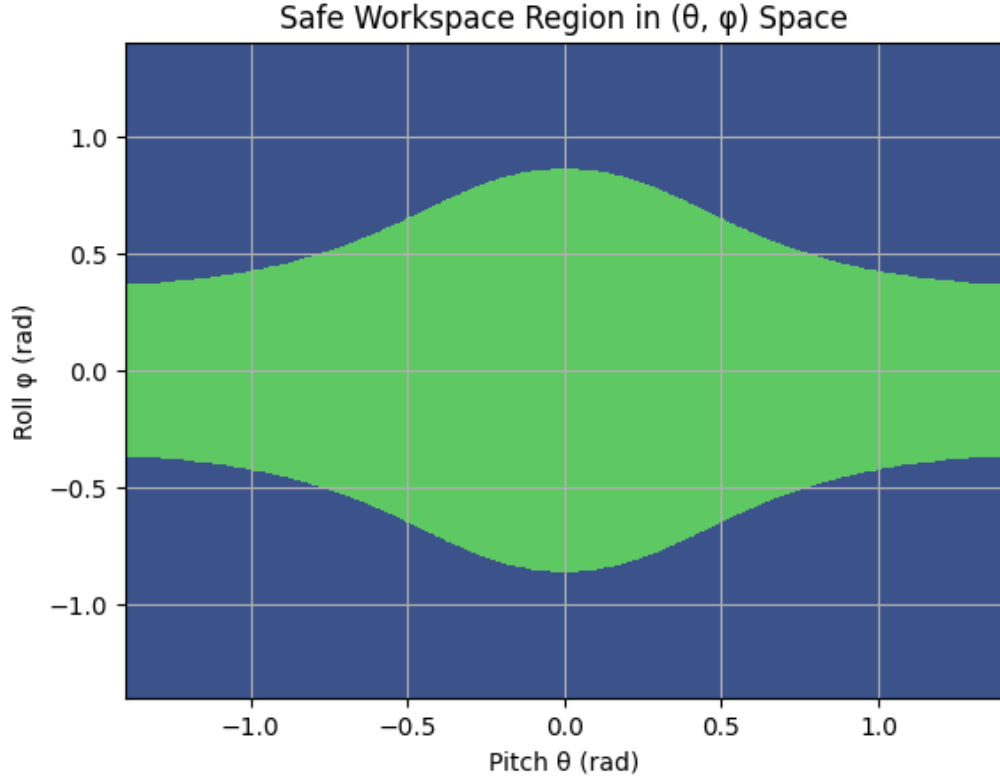


Figure 3: Actuator displacements (Z_1, Z_2, Z_3) for Case 3: Combined Motion.

That is:

- Leg 3 becomes horizontal
- Square-root term becomes zero
- Jacobian denominator blows up

These are true geometric singularities.

Second Plot — Safe Workspace Region

The green region satisfies:

$$G > 0 \quad \text{and} \quad \cos \theta \neq 0$$

Meaning:

- All legs have valid length
- No horizontal alignment
- Jacobian determinant $\neq 0$

This region is your:

Singularity-free orientation workspace

— Start From Singularity Condition

From earlier, a singularity occurs when the square-root term becomes zero:

$$G = 0$$

Where:

$$G = L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2$$

So the boundary equation is:

$$L^2 = (S_m \sin \theta \sin \phi)^2 + (S_m \cos \phi - S_{B3})^2$$

This is your workspace boundary surface.

— Expand the Equation

Expand both terms carefully.

First term:

$$(S_m \sin \theta \sin \phi)^2 = S_m^2 \sin^2 \theta \sin^2 \phi$$

Second term:

$$(S_m \cos \phi - S_{B3})^2 = S_m^2 \cos^2 \phi - 2S_m S_{B3} \cos \phi + S_{B3}^2$$

— Substitute Back

$$L^2 = S_m^2 \sin^2 \theta \sin^2 \phi + S_m^2 \cos^2 \phi - 2S_m S_{B3} \cos \phi + S_{B3}^2$$

— Rearrange

Group the S_m^2 terms:

$$L^2 = S_m^2 (\sin^2 \theta \sin^2 \phi + \cos^2 \phi) - 2S_m S_{B3} \cos \phi + S_{B3}^2$$

Now simplify the bracket using trigonometric identities:

$$\begin{aligned} \sin^2 \theta \sin^2 \phi + \cos^2 \phi &= \cos^2 \phi + \sin^2 \theta \sin^2 \phi \\ &= (1 - \sin^2 \phi) + \sin^2 \phi \sin^2 \theta \\ &= 1 - \sin^2 \phi (1 - \sin^2 \theta) \\ &= 1 - \sin^2 \phi \cos^2 \theta \end{aligned}$$

A beautiful simplification.

— Final Boundary Equation

Substitute the simplified bracket back into the main equation:

$$L^2 = S_m^2 (1 - \sin^2 \phi \cos^2 \theta) - 2S_m S_{B3} \cos \phi + S_{B3}^2$$

Rearrange to isolate the coupled term:

$$S_m^2 \sin^2 \phi \cos^2 \theta = S_m^2 + S_{B3}^2 - 2S_m S_{B3} \cos \phi - L^2$$

This is your exact singularity boundary surface equation.

— **Express ϕ as a Function of θ (Boundary Curve)**

Rewrite as:

$$S_m^2 \cos^2 \theta \sin^2 \phi = S_m^2 + S_{B3}^2 - L^2 - 2S_m S_{B3} \cos \phi$$

This is a nonlinear implicit relation between θ and ϕ . You cannot isolate ϕ in a simple closed form—but this equation itself defines the exact boundary.

— **Special Case Insight (Very Important)**

Case $\theta = 0$:

Then $\cos^2 \theta = 1$. The boundary reduces to:

$$S_m^2 \sin^2 \phi = S_m^2 + S_{B3}^2 - L^2 - 2S_m S_{B3} \cos \phi$$

This simplifies to a quadratic equation in terms of $\cos \phi$. This mathematically explains why your safe region has its maximum roll capability exactly at $\theta = 0$.

Physical Interpretation

The boundary equation describes:

- The exact condition where leg 3 becomes horizontal.
- The limit beyond which a real solution for Z_3 does not exist.
- The point where the workspace collapses.

The curved shape in your plot comes directly from the coupled $\sin^2 \phi \cos^2 \theta$ term, which links pitch and roll.

The orientation workspace boundary is obtained by equating the Jacobian denominator to zero, leading to the implicit relation $L^2 = (S_m \sin \theta \sin \phi)^2 + (S_m \cos \phi - S_{B3})^2$. This equation defines the exact singularity surface separating feasible and infeasible configurations.